

Compounding with Different Time Intervals



Interest can be compounded at different time intervals, as well as annually. It could be compounded semi-annually (every 6 months), quarterly (every 3 months), monthly, daily or at any other time interval.

- 1). a). \$100 is invested. Calculate the total amount of money saved after 1 year if the interest is 12% p.a., and interest added at the end of the year.
 b). Does it matter if you calculate this using simple interest or compound interest?
 c). \$100 is invested at a rate of 12% p.a. compounded every month for 1 year.
 Note: The annual interest rate of 12% p.a. is divided by 12 for the monthly interest rate.
 Note: Do not round your answers until the end.
 Copy and complete the table.

End of	month 1	month 2	month 3	month 4	month 5	month 6
Total						

End of	month 7	month 8	month 9	month 10	month 11	month 12
Total						

- d). Use the compound interest formula to determine the rate of interest p.a. that would give \$112.68 at the end of one year after investing \$100 initially.
- 2). a). \$100 is invested at a rate of 12% p.a. compounded every two months for 1 year.
 Note: The annual interest rate of 12% is divided by 6 for the monthly interest rate.
 Copy and complete the table.

End of	month 2	month 4	month 6	month 8	month 10	month 12
Total						

- b). Use the compound interest formula to determine the rate of interest p.a. that would give \$112.62 at the end of one year after investing \$100 initially.

The answers to 1).d). and 2).b). are called the **effective annual interest rate**, whereas the quoted interest rate of 12% p.a. compounded monthly is called the **nominal rate**.

A more complex formula can be used can be used,

$$\text{final amount, } A = P(1 + \frac{r}{n})^{nT},$$

where P - Principal amount, r - depreciation rate as a decimal, T - time in years, n - number of times the interest is compounded per year.

Example \$400 is invested at a rate of 8% p.a. compounded daily for 2 years.
 What is the total amount at the end of the 2 year period?

$$A = P(1 + \frac{r}{n})^{nT}, \quad A = 400(1 + \frac{0.08}{365})^{365 \times 2} \quad A = \$469.40 \quad (2 \text{ d.p.})$$



Find the total amount for the following. Leave answers to 2 decimal places.

	Principal, P	Interest rate p.a., r	No of times compounded in a year, n	Time in years, T		Principal, P	Interest rate p.a., r	No of times compounded in a year, n	Time in years, T
3).	\$365	14%	2	4	8).	\$543.40	4.5%	24	6
4).	\$1 438	7%	4	3	9).	\$45.84	0.6%	365	10
5).	\$6 584	3%	3	5	10).	\$4 734.50	7.3%	36	5
6).	\$74	24%	6	4	11).	\$473.71	17.4%	52	8
7).	\$354	9%	12	5	12).	\$9 036.83	3.2%	365	12